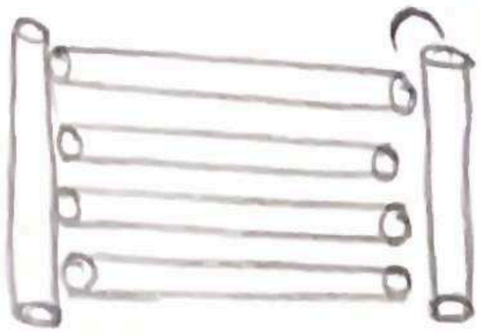
27.02.2016
(Friday Reflection)

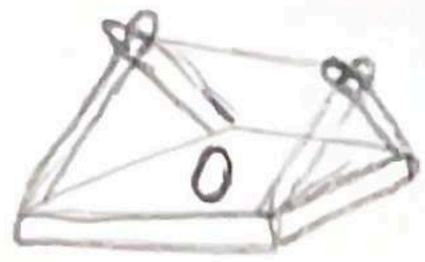
My First Prototype Reflection of the Egg Drop Experiment:

- First I tried to make a case for the egg. Since we had only two materials I thought about our previous work about Robison Crusoe. So I made a case like a float.



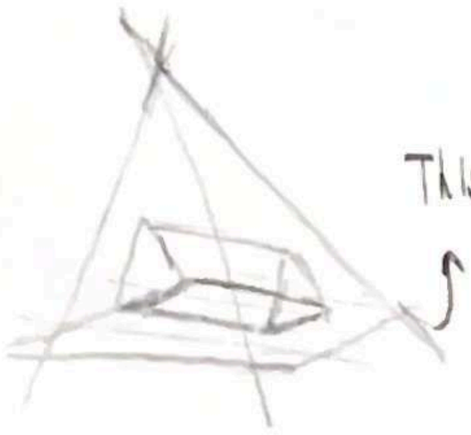
→ connected everything with rope.

→ Nest



! Rope was slippery and it teared down on its own. This structure was not holding the egg properly and needed some other structer to be in.

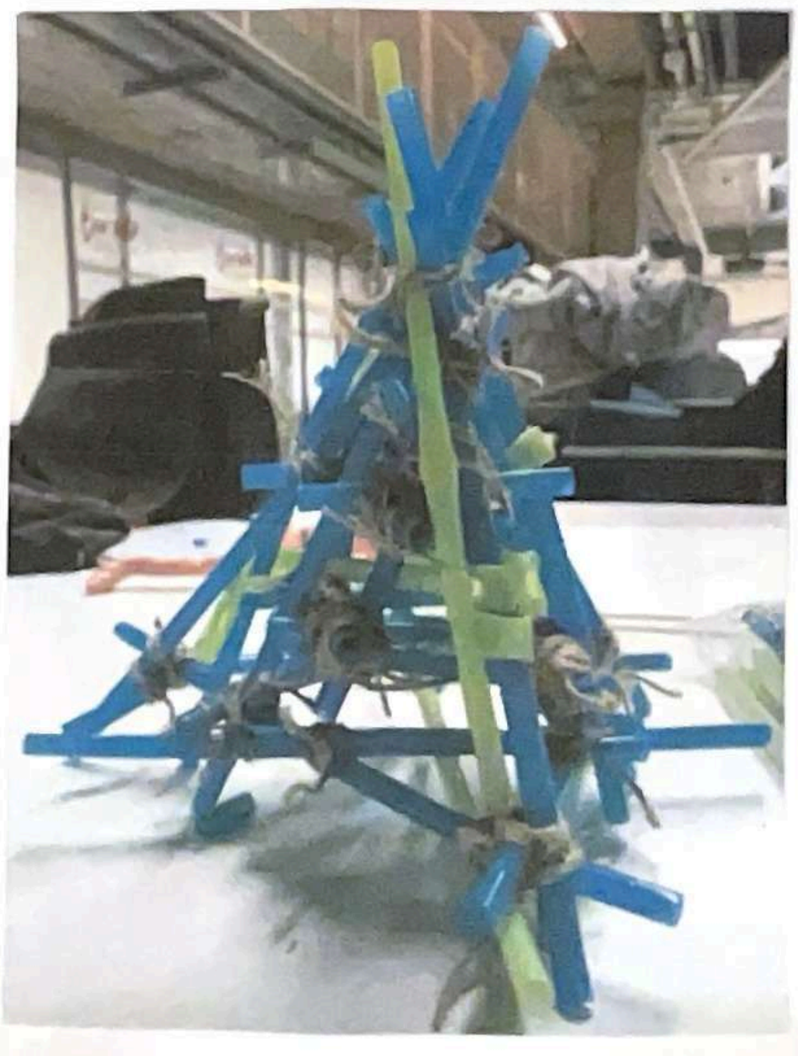
- Then I decided to cover the float I made with a triangle base and put the float inside. (Almost like a tent) My problem was that I was not satisfaid with the empty spaces left and the structure didn't seem well. So I kept adding to straws.



This outer parts were supposed to help with the fall while easing the fall, absorbing.

→ These outer parts were supposed to help with the fall. They did help but because of the stiffness of my design and the way they placed randomly it didn't help like I wanted.

→ The straws I used didn't have accordions so they had problems with bending and exploded more easily.



→ The design looks so random and because I focused on strenght rather than focusing on stability and streching. I kept the knots too stiff and too much which made my design heavier. This is why I believe the restrons on the materials will be helpful for the iteration of my design. I am going to focus on stability with less product and give attention to streching while giving the egg some space to absorb the force.

→ There were too many random straws placed in order to keep the egg still. This problem didn't give any space to the egg and it directed the force to the egg.

→ There are lots of breakage on the straws as well and they worked as a breaking point for the straws. It weakened the design.

Working With AI : Chatgpt

28.02.2026
(For Tuesday)
(week 3)

What went wrong:

- Too much straws to make it "strong" but strenght is not the goal → energy absorption is.
- Force was not redirected.
↳ Shock concentrated at the bottom.
↳ My prototypes base looks <u>stiff</u> and <u>flat</u>, meaning the impact force went straight up into the egg.

Must Have:

- Crumple zone
- Suspension stretch
- Energy delay system
- The more tight knots = the more the structure acts like a solid object.

To Save The Egg

- Increase the time of impact.
- Spread force over distance
- Convert sudden force into gradual absorption.

Concept Ideas:

1- Instead of rigid pyramid → light cube frame
Then suspend the egg in the center using rope like a trampoline.

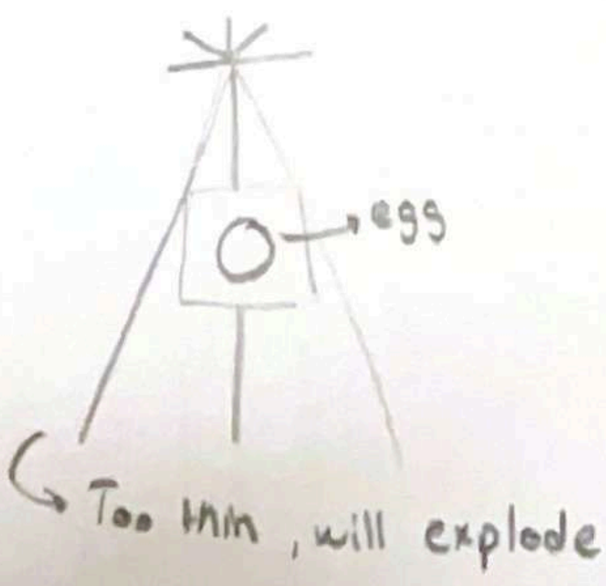
} The frame absorbs first impact.
The rope stretches
The egg moves slightly
Force is delayed.

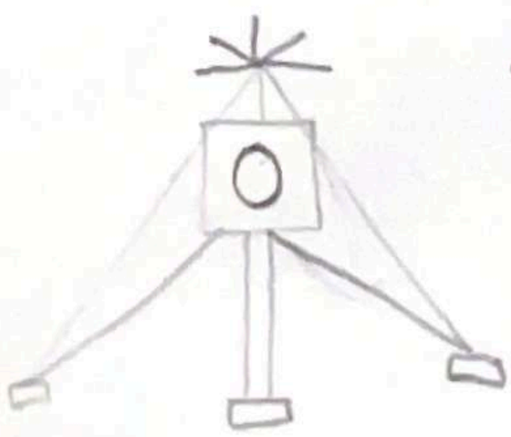
2- Helicopter Landing Gear

Key Points For The Next Design:

- Fewer joints
- Let things bend
- Egg must not touch the frame
- Design for bottom impact → assure it lands bottom first.

- Helicopter Landing Gear + A star shape Hat





=> The egg is not secure with the ropes = another way to hang the egg:

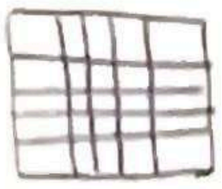
→ might be a better option

Working With AI : Chatgpt

28.02.2026
(For Tuesday)
week 3

Better Rope Suspension Options:

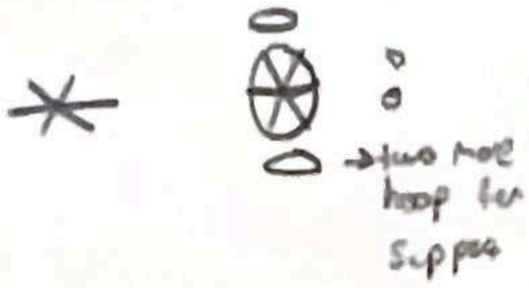
- Rope Basket



=> make a nest

- Point Sling

- Cross + Vertical Safety Loop

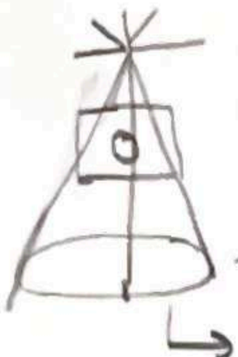


while testing => Does the ess:
- swing violently
- Slam into frame
- Shift to one side

↳ if yes increase symmetry, not tension

- Hammock Ring

The problem with the legs:



→ will explode

↳ + adding a bottom ring is the best option.

- add more straws to the legs
- bend the legs = absorbs force easily = but the straw would break anyways.
- knee joint = rope = not good for the weight
  - might not hold.

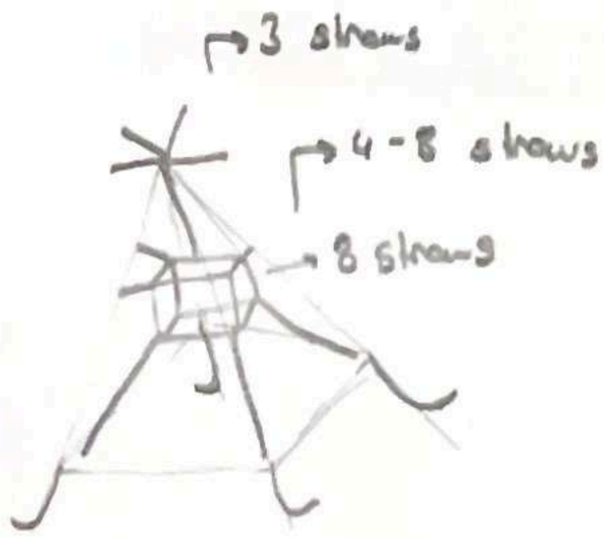
In Addition:

I really liked the tetrahedral pyramid idea Gemini gave me and I think these two final products deserves to be combined. Since I decided to add a bottom ring to the Chatgpt's design I think the ring could be rectangular and I can combine the structures together.

- It is the most stable 3D shape => not too rigid → flexibility

↳ micro crumple legs:

- extend each leg 1-2cm below the ring = increases stopping the



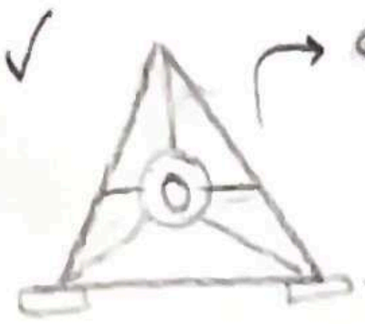
Working with AI: Gemini

28.02.2026
(For Tuesday)
(week 3)

- My structure appeared strong, yet it was too flexible, it couldn't absorb the energy of the impact.
- too rigid → directly transfered to the egg!
- too much rope and straws = heavy = more kinetic energy
- non-absorbent connection = loose, bulky lashing, ropes didn't provide a secre.
  ↳ seat belt + an airbag → solution to crashing => thight hold of ropes!!
- egg must be suspended = allows it to move slightly
- lighter!
- helping the design fall in a certain way.

1-st ideas with AI:

✓



→ egg is at the center but loose

→ crumple zone = helping How the product must fall

someone in the class made it the thinking is good but when it falls side ways the egg cracks.
X

cage

→ egg still at the center

→ showing how to fall

→ Hybrid Stability Cage = too much straw is need x
↳ Simplification and Scaling => Tetrahedral - w Cage

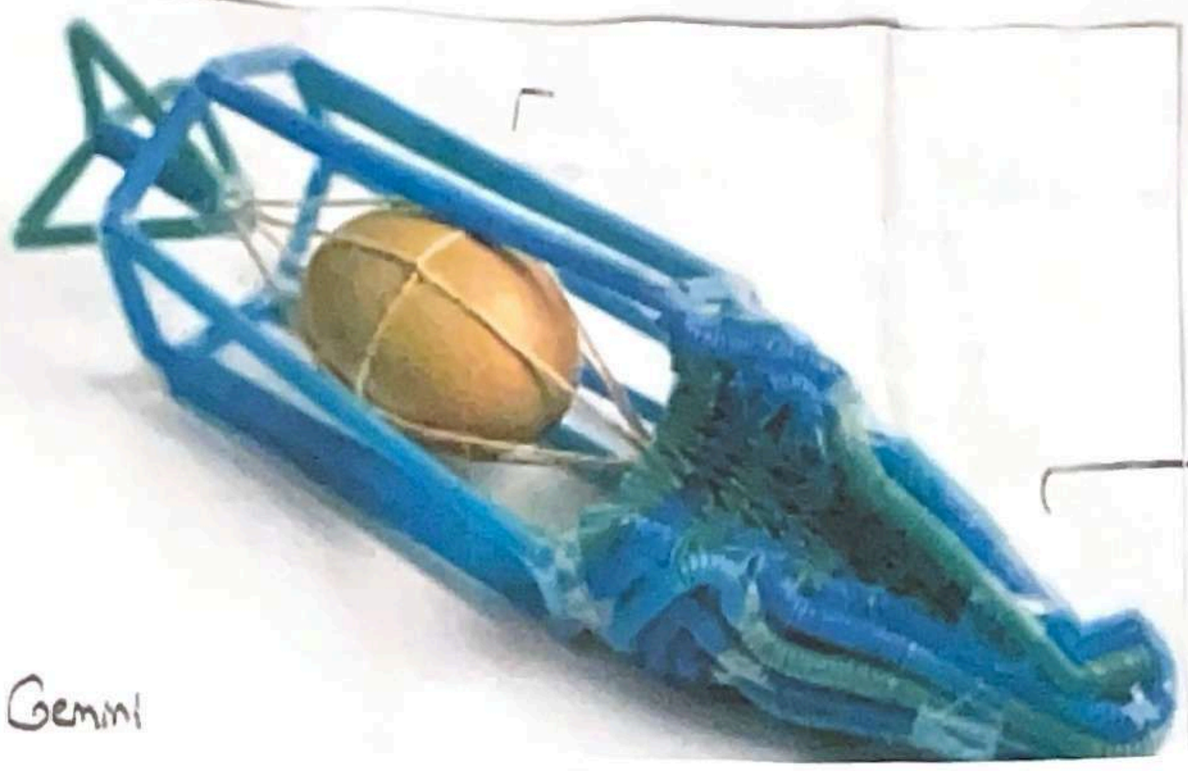
→ Use the accordion parts of the straws to help with exploding.
→ Triangle pyramid = hard to tip over because it is stable.

Working With AI : Combined

01.03.2026
(For Tuesday)
(week 3)

①

- I started with talking about the problems my first design has and focused on how to improve them.
- Some of the pictures and designs was not real or doable. This is why it was easier to communicate with chatgpt and getting help from Gemini sometimes helped me the most.

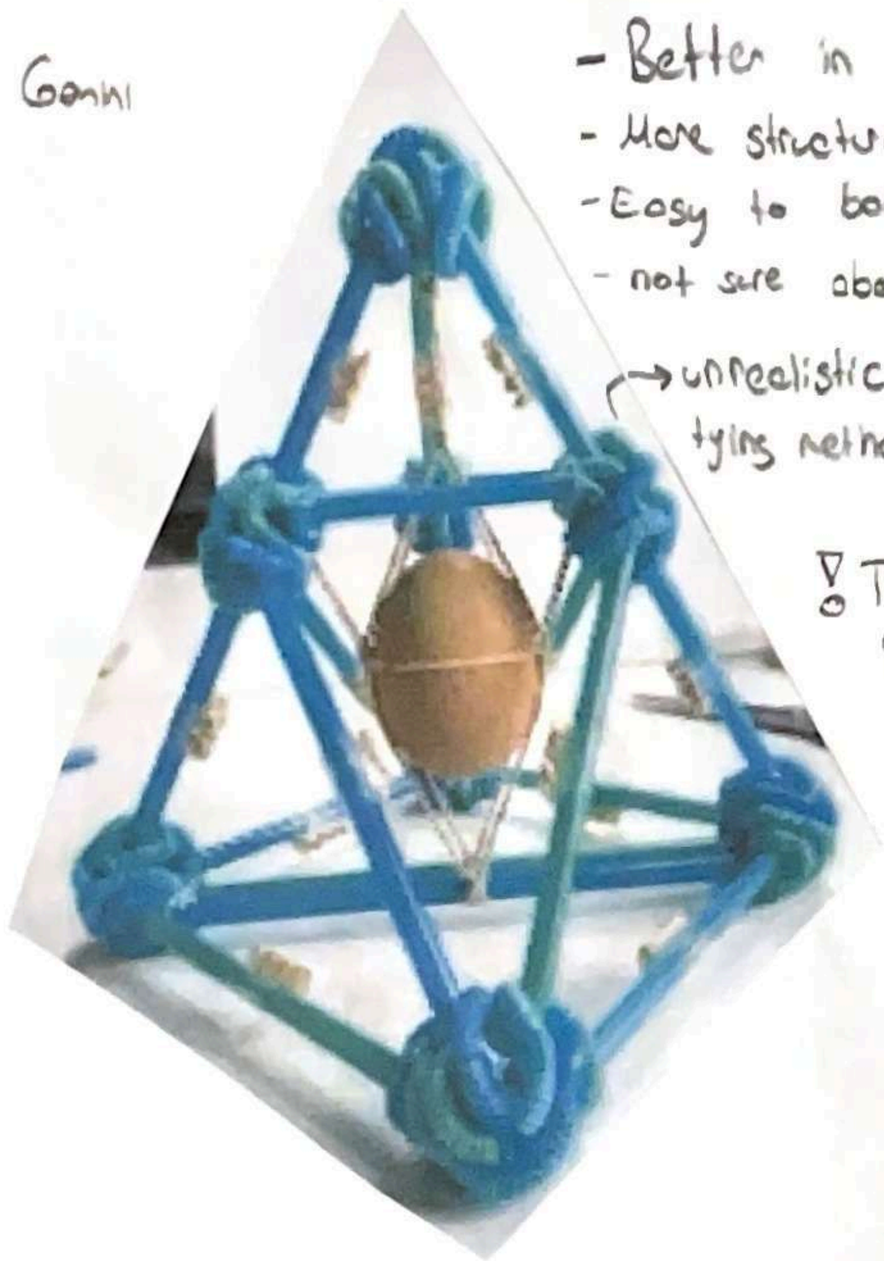


Gemini

- Has a high chance to fall to the side.
- The crumple-point idea is good but not like this picture.
- The top idea is good in order to help with the air resistence. However rather than this unrealistic design we could upgrade the air resistence.

→ unrealistic (doesn't know how to connect the straws)

Gemini

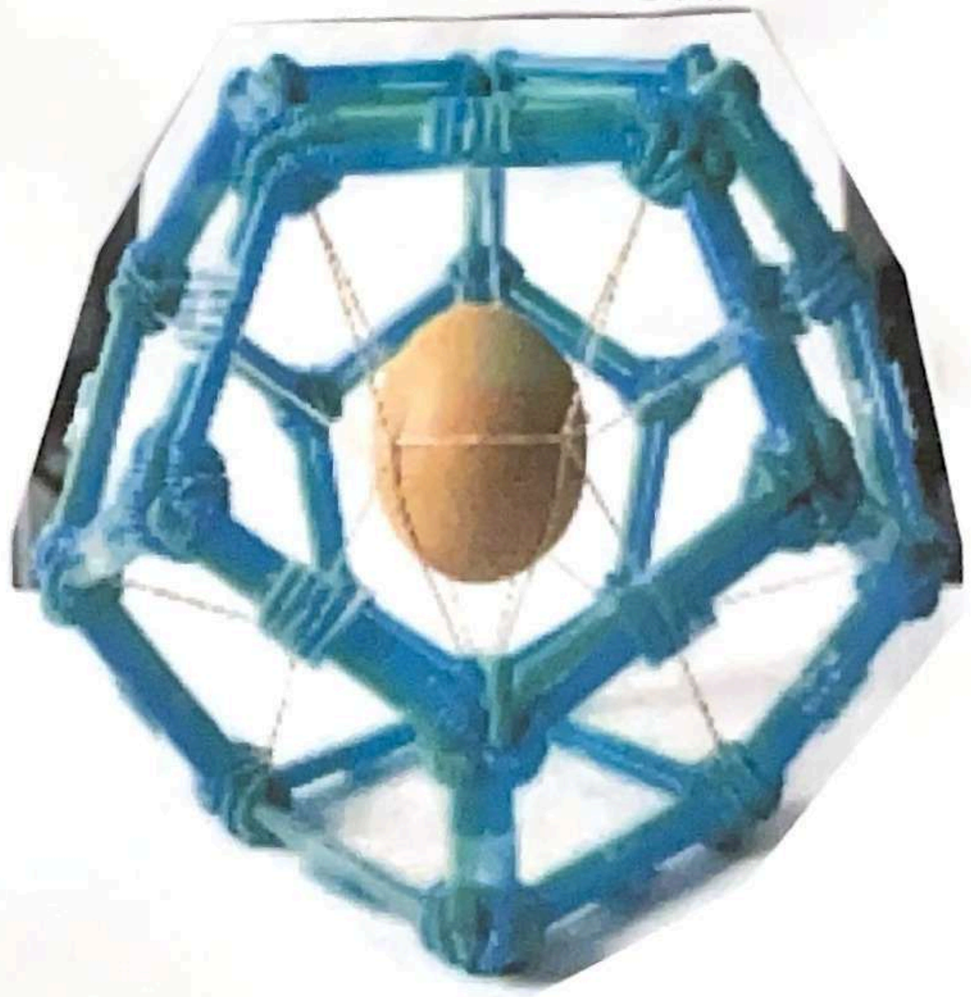


- Better in shape.
- More structured.
- Easy to balance → too strong, heavy
- not sure about the way the ess stands

→ unrealistic tying methods

! Tetrahedral Pyramid

Gemini



- Really good and accepteble design
- However can not be achieved with the 20 straws max limit.

- After working with Gemini I had some ideas in my mind. Gemini's drawings and thinking system was really hard to understand because it was not realistic and hard to control. So I had to re imagine everything it gave me once more for accuracy and redesign it again.

- This is why I switched to Chatgpt. Both for checking Gemini's answers and trying to look from an another perspective. Because Chat is more mathematical rather than Gemini or it feels like it.

Working With AI: Combined

01.03.2026
(For Tuesday)
Week 3

- I got feedback from Chat GPT as well and find out what my mistakes were. I also compared the answers with Gemini as well to see in different perspectives and check if they were true.

The final essentials looked like this;

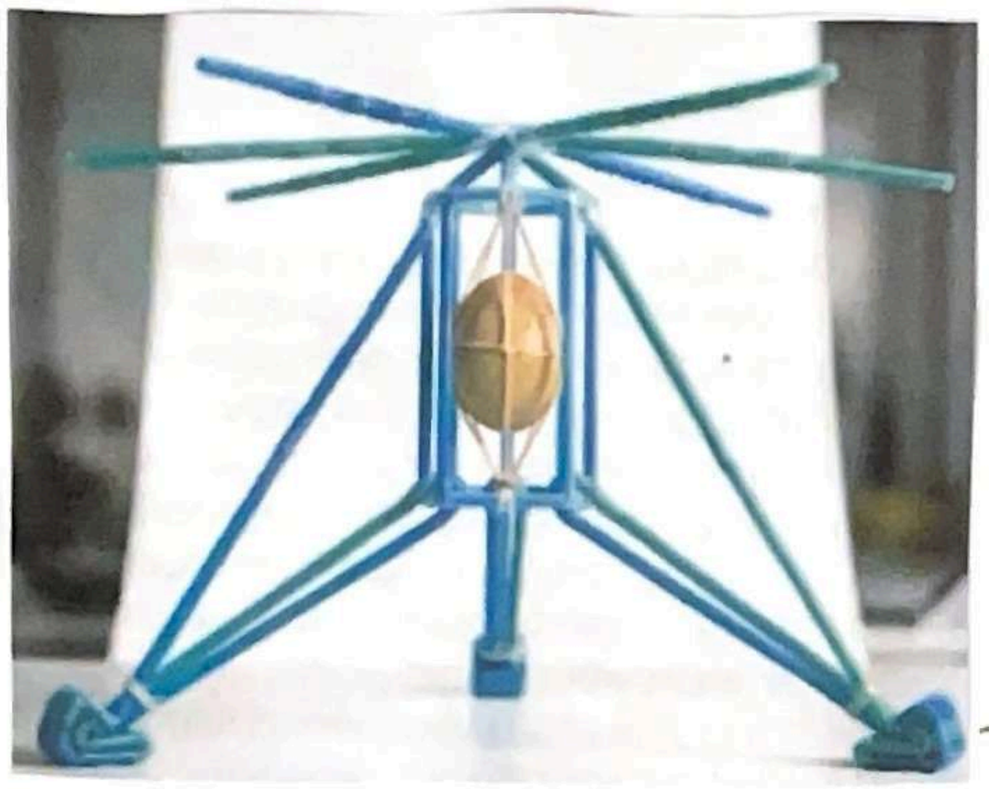
Must Have:

- Crumple Zone = for the bottom of the design
- Suspension Streeth = for the ess
- Energy delay system = top part for air resistance

To Save The Ess:

- Increase the time of impact
- Spread force over distance
- Convert sudden force into grounded absorption.

Helicopter Landing Gear Idea:



Gemini

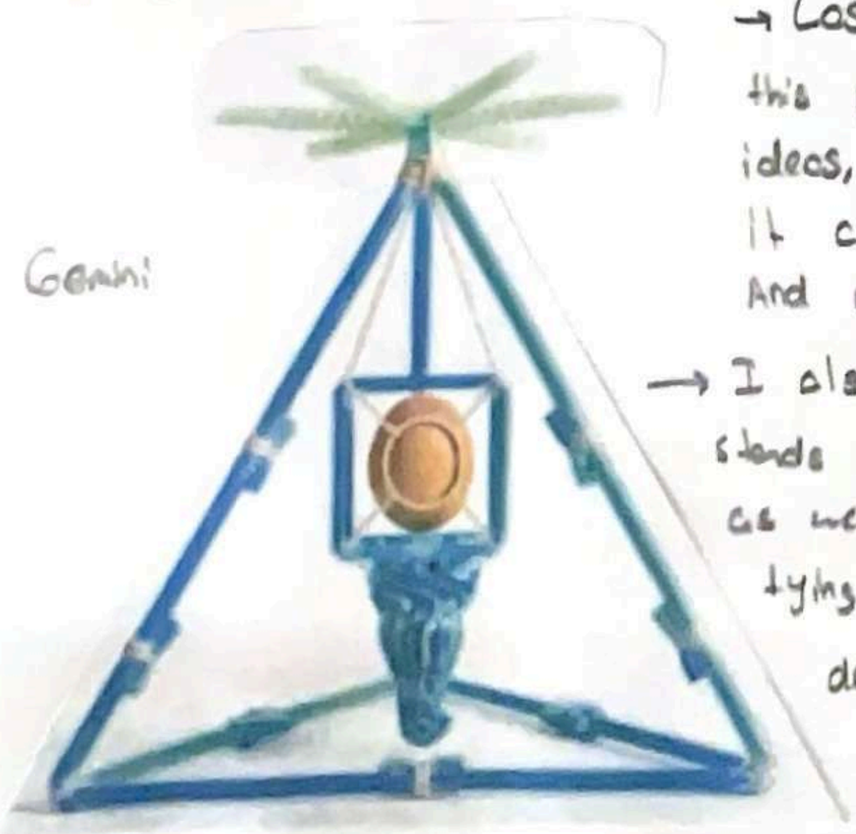
→ top part is helpful for air resistance

→ The cage for the egg is a good idea

→ connections are not realistic = must find an another way for the legs because they might explode.

→ crumple-zone = not realistic the way it has been made

- So I decided to combine these two ideas into one. The last Chat GPT picture has pottential but not stable and might not work. The first picture is not impossible to do with straws but has a good structure.



Gemini

→ Lastly Gemini gave me this picture with the combined ideas, Chat suggested that it could be a tetrahedral pyramid And must have little crumple points.

→ I also didn't like the way the ess stands there, it is not realistic as well. So I asked ~~for~~ some tying ideas to Chat and decided on one.

Chat GPT

↳ Ess is not scured

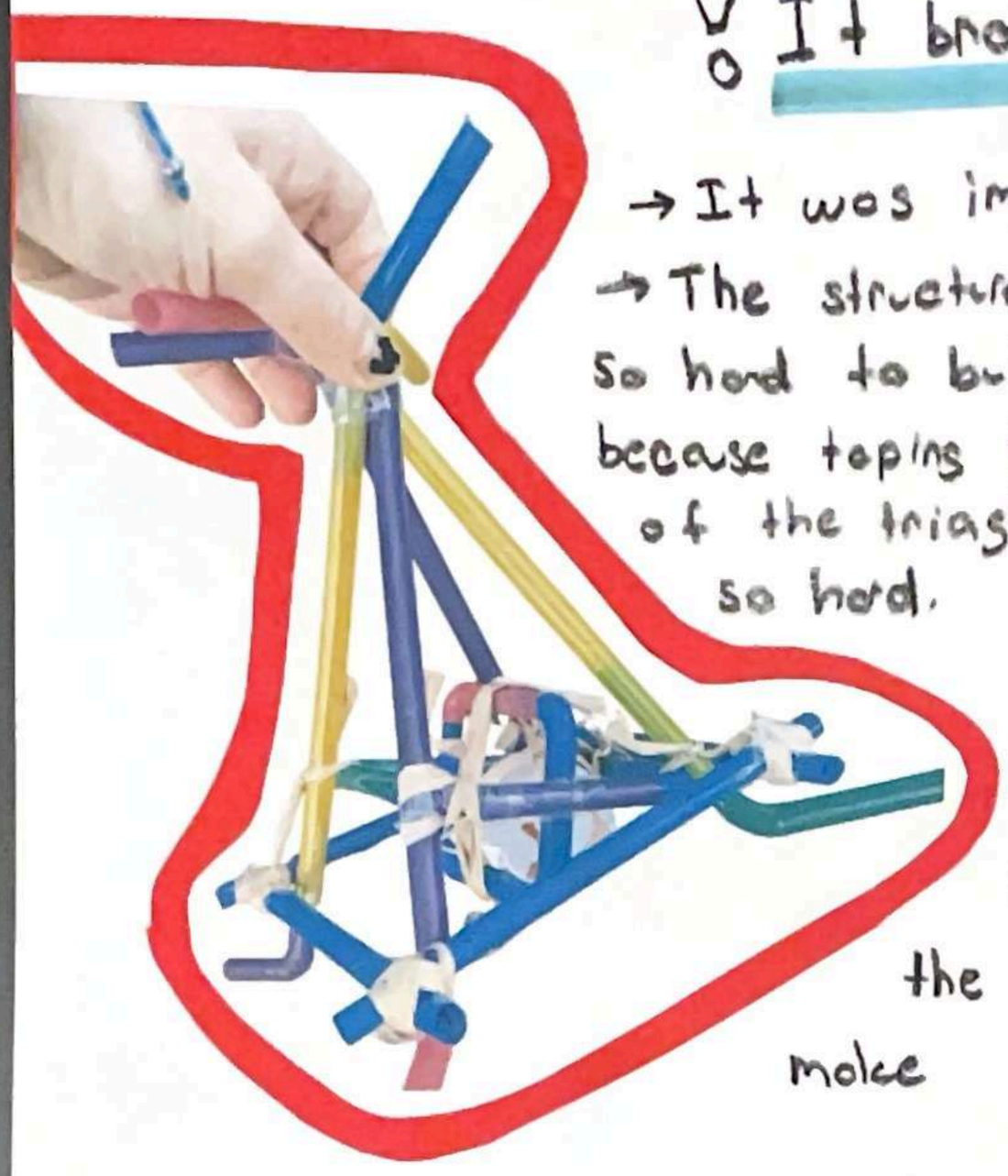
↳ Less are not supported

↳ less are too thin = needs a bottom ring to help not to break

→ Basically my design process took help from AI but didn't or couldn't copy the exact picture, design they gave because most of them were not realistic and needed iteration from a person. This is why I asked for lots of ideas and choose their best sides in order to collabrate them.

First Prototype Review:

! It broke !



→ It was imbalanced.
→ The structure was so hard to build because taping the top of the triagle was so hard.

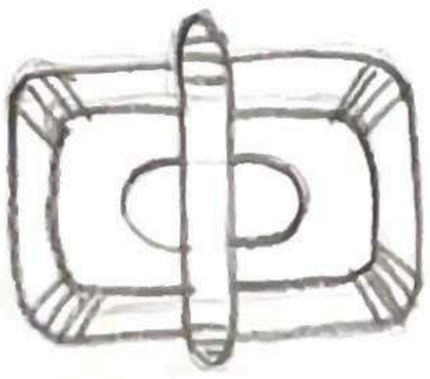
→ Top part was unnecessary, 4 straws were not enough to create air-resistence.

→ Legs were too small to create crumple-zone.

↳ The nest we created with AI was so hard to make with little pieces of straws.

→ I discovered I could intertwine the straws into each other. This technique diccreased the use of tape and rope and helped to make the structure less heavy.

→ It was so hard to work with an egg. It was slippery and I got scared that It might explode. So I couldn't create a secured nest for the egg.



=> Rather than the nest we designed with AI I was able to make this one. The egg didn't sit inside well and attaching this to the structure was harder than I thought.

→ AI couldn't think building wise so I struggled a lot.

→ The nest was closer to the ground and this made the egg crack easily + the crumple points were too small and bended too much it was not helpful. It was too shallow and didn't stretch well.

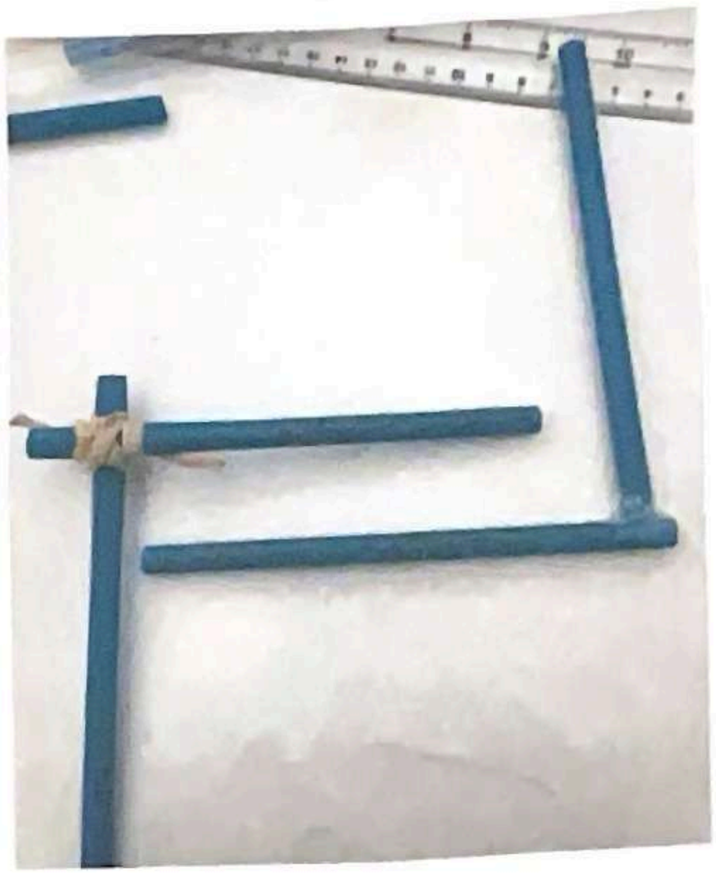
∝ So, I decided to ask for something easier to build with straws. Focusing on the material for AI to understand. I specify the mesurments again to create a better structure.

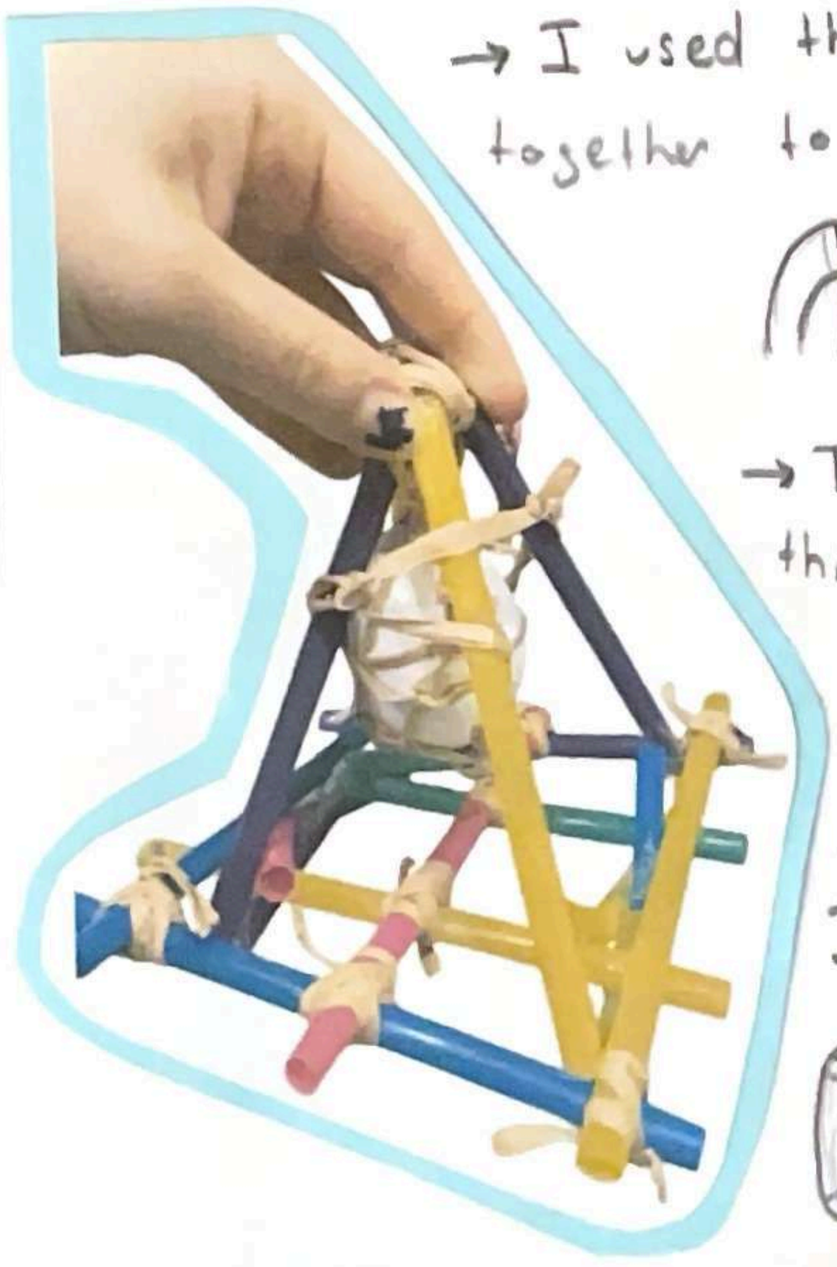
→ The top was unneciery.
→ Nest was so hard to build and attach to the base.
→ The crumple-zone was not enough.
→ The structure was hard to build.
→ Needs more balance so it wouldn't flip over.

Second Prototype Review:

→ I decided to use a prop egg to help me build around it more easily.
→ I didn't use the idea of making a nest from the straws and focused on making a hammock for the egg.
→ I did knots around it in order to create a bag. It was more easy to work with and this design easier to attach.

→ I changed the ring at the bottom to a rectangle and added 2 extra straws to prevent the egg from touching the ground.
→ I attached the upper parts to the bottom (the accordion pieces) in order to create triangle-zones.



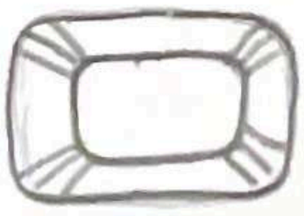


→ I used the way of interwining the straws together to do the tip of the pyramid.

= then tied it together to keep it strong.

→ I tried different tying methods.
→ I used tape for a looser structure.
→ I used rope for a more stiff structure.

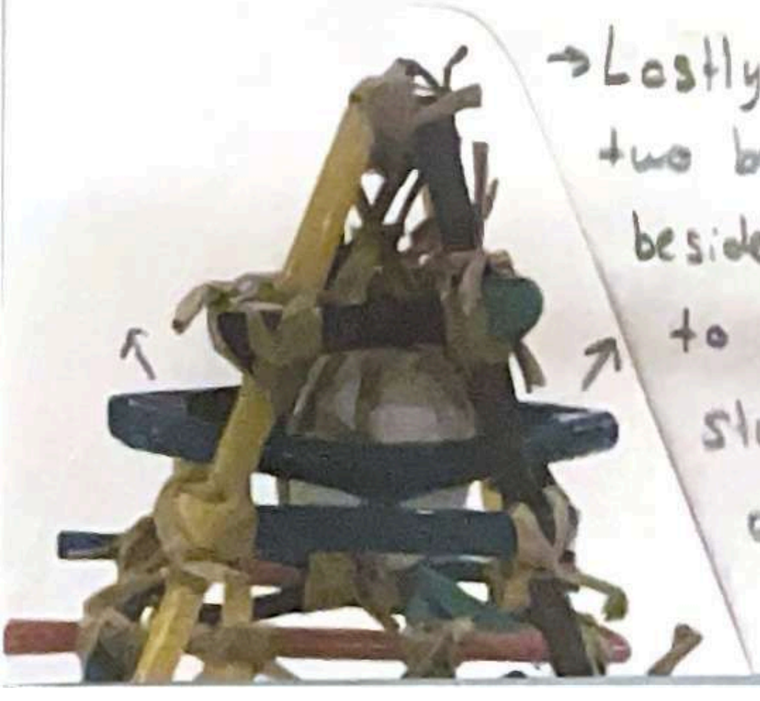
→ This was the first prototype of this design but it did fall to the side once.
So I had to make a protection for that as well.
ChatGPT has been suggesting to add rings to the sides of the design from the start of the project.
I added two more rectangular rings to the outer corners and attached them with rope



Rings

=> I used the same technique I did before.

∝ This was the final design I made with the help of AI.



→ Lastly I added two bendable straws beside the egg to give some streching points and strenght.